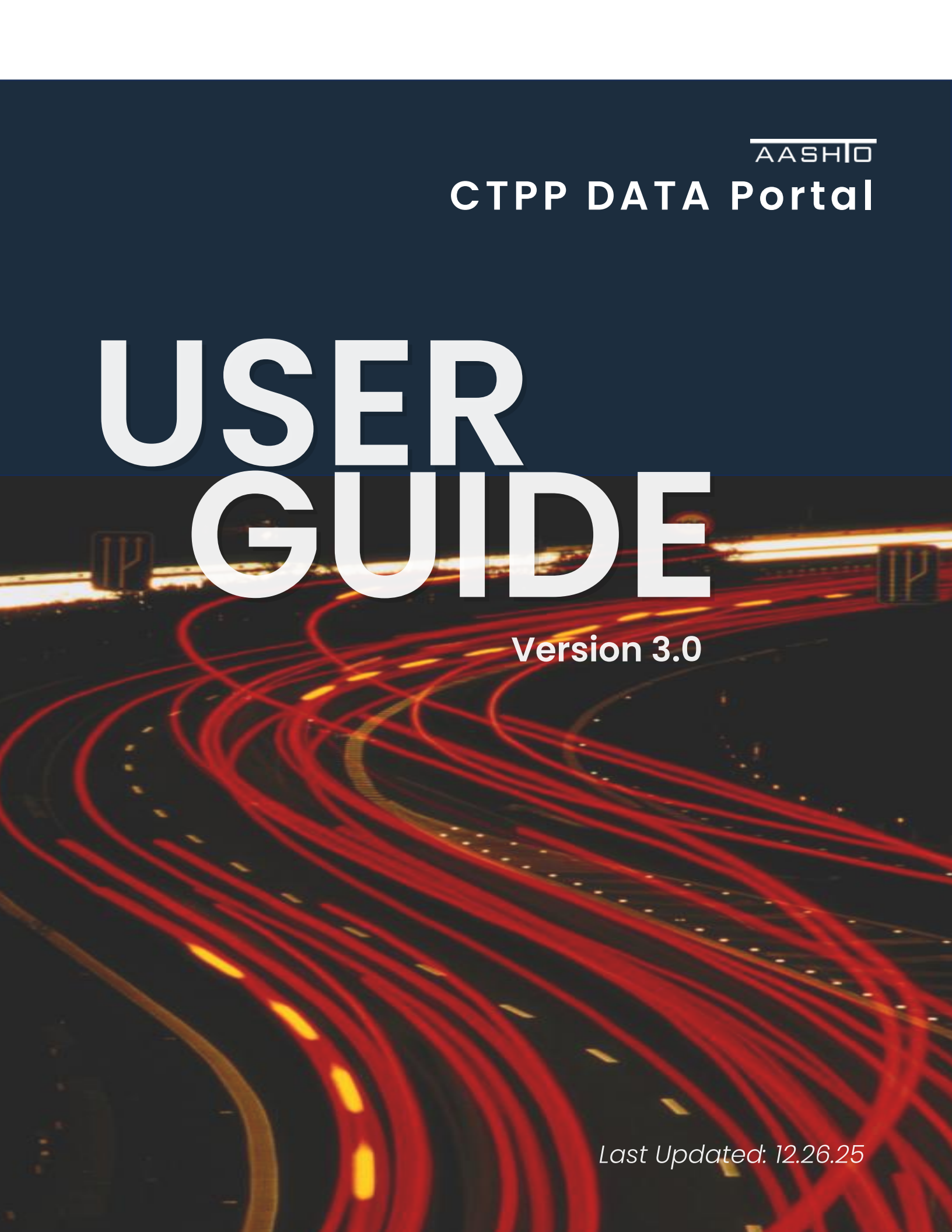




AASHTO

CTPP DATA Portal

USER GUIDE

Version 3.0

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1. Portal Overview

Core Functionality

The new [AASHTO Census Transportation Planning Products \(CTPP\) Data Portal](#) offers users new and improved functions in the following areas:

- Data query
- Query result display and customization
- Query result visualization
- Data portal general utilities
- API portal

General Features

Many user-friendly features have been built-in to make the portal intuitive and easy to navigate, including those shown in Table 1.

Table 1. Key CTPP Data Portal Features

Feature	Technical Description
+	Expand. This symbol indicates that its associated item is expandable. Select this symbol to expand and display either a sub-list or more information.
–	Collapse. This symbol indicates that its associated item is collapsible. Click this symbol to collapse the sub-lists or detailed information. Collapsing a sub-list will not deselect the selections in the sub-list.
□	Select checkbox (option). This symbol indicates that its associated item can be selected. Select to include the item and select again to remove the item. This symbol also indicates that multiple selections can be made.
○	Select radio button (required). This symbol indicates that its associated item is one of a predefined set of mutually exclusive options. The user can choose one of the options by selecting its radio button.
▼	Select dropdown. This symbol indicates a dropdown list is available. Click to display the dropdown list or more information. Click again to collapse the dropdown list.
▼ ▲	Scroll vertically. These symbols indicate that the visible content is just part of the whole content. The user can select these symbols to scroll up and down to see other parts of the content.
◀ ▶	Scroll horizontally. These symbols indicate that the visible content is just part of the whole content. The user can select these symbols to scroll left and right to see other parts of the content.
⌵ ⌶	Collapse/expand. These symbols indicate that the header area of the portal can be collapsed or expanded.
Deselect all	Deselect all. Selecting this link will deselect all selections made in the hierarchical list.
Select on map	Select on map. This option gives the user a map to select geographies of interest using visual boundaries.

2. Data Query

Data Query Interface

The CTPP Data Portal homepage provides an intuitive interface for users to query CTPP data:

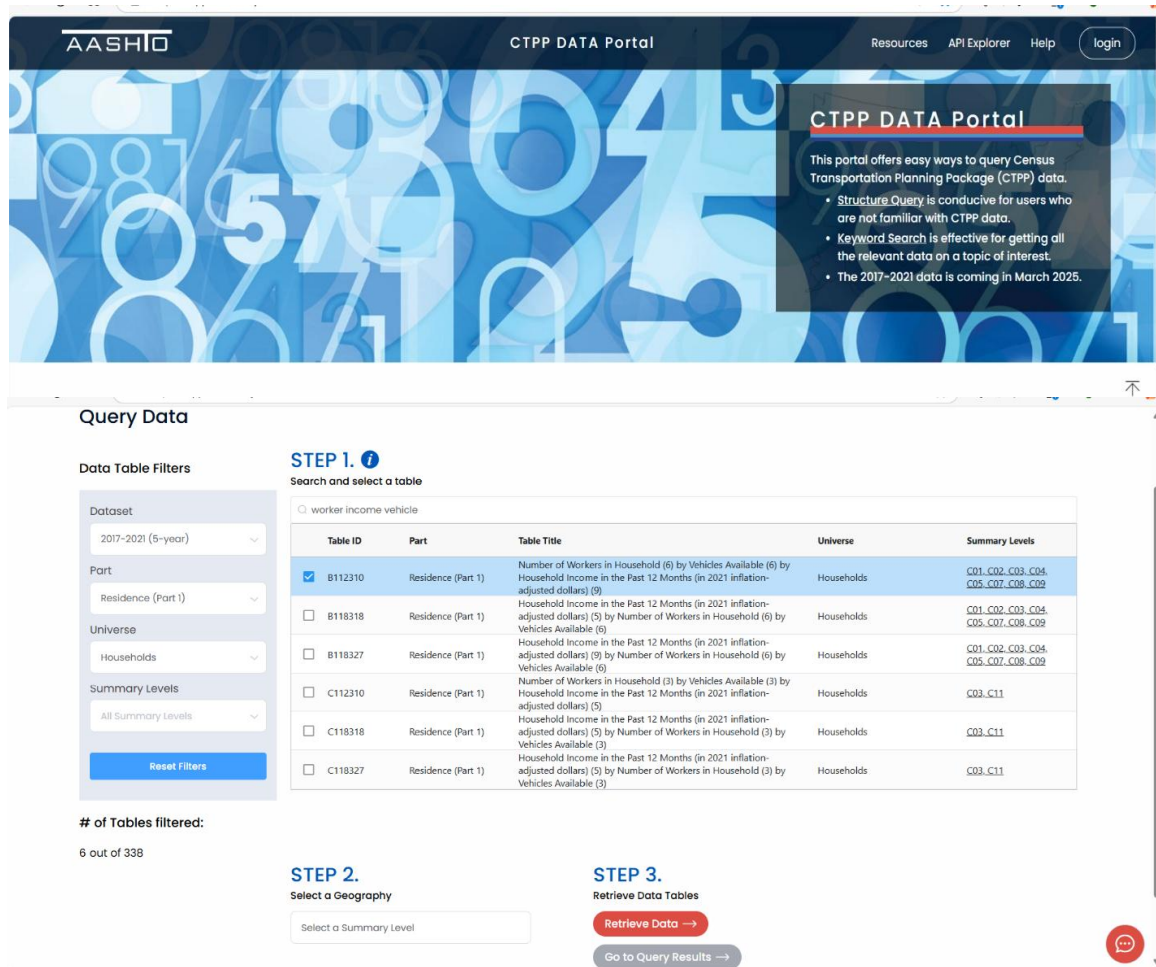


Figure 1. CTPP Data Portal Homepage

The **Query Data** interface contains four Data Table Filters and one Keyword Search function.

- **Dataset Filter:** The Dataset Filter allows users to select CTPP data tables by the period during which the data were collected. A CTPP dataset contains all the data tables that are generated from the data collected during the same period. Currently, the CTPP Data Portal contains the following five datasets:

- 2000 Dataset
- 2006–2010 Dataset
- 2012–2016 Dataset
- 2017–2021 Dataset

The default value of the Dataset Filter is the 2017–2021 dataset.

- **Part Filter:** The Part Filter allows users to select CTPP data tables by Residence (Part 1), Workplace (Part 2), and Flow (Part 3).
 - The Residence tables focus on the characteristics of the population from a perspective of where people live. It provides data about the residents of specific areas, such as their socioeconomic characteristics (age, ethnicity, education, occupation, income, etc.) and their housing situations.
 - The Workplace tables contain data on where people work. They provide details about employment locations, including the type of industries people work in and their employment status.
 - The FLOW tables provide data that links the Residence and Workplace. It captures the commuting patterns between where people live and where they work. This part of the data shows the flow of commuters from their Residences to their workplaces, indicating the travel routes, modes of transportation, and distances involved. Flow data is one directional, always from Residence to Workplace.
- **Universe Filter:** The Universe Filter allows users to select CTPP data tables by different population groups. "Universe" in CTPP means the entire group of individuals, households, or workers that are included in a specific data table or represented by the data in that table. For example, if a CTPP table is focused on population 16 years and over, the "Universe" would represent all people who are 16 years old or older. Since CTPP data were collected with different focuses or special interests during different time periods, the universes of different CTPP datasets may be different.
- **Summary Level Filter:** The Summary Level Filter allows users to select CTPP data tables by the level of aggregation at which the data is presented or, in other words, the geographic level of the data aggregation. Summary level indicates how detailed or broad the dataset is in terms of geographic area. Summary levels are used to categorize data by different geographic boundaries and allow planners and analysts to examine patterns at various spatial scales.

- **Keyword Search:** The Keyword Search function allows users to find CTPP data tables by matching a user's search query (keywords) to the contents of CTPP data tables. While Keyword search and filtering are both methods used to find relevant information, they work in different ways. Filtering drills down into a set of results or narrowing choices by predefined attributes of the searched items. Keyword search finds relevant items by matching the content of the item to the keywords the user used to search. Keywords are not predefined and can be anything relevant to items the user intends to find.

The Keyword Search function of the CTPP data query has the following built-in features:

- The scope of the keyword search is the filtered results or narrowed choices from filtering.
- Fuzzy search technique is used to find matches that are close variations of or similar to a given search term, rather than requiring an exact match. For example, a search for "income" and a search for "earning" may generate the same result.
- Boolean operators "AND" is used for searches with multiple keywords. This means the search will try to find only those items whose content contains all keywords entered for the search. Consequently, entering more keywords will generate a shorter list of items found. For example, a search with "worker income vehicle" will generate a list of 3-dimensional CTPP tables by worker, by income, and by vehicle.

In addition to the Filters and the Keyword Search, the Data Query Interface also provides information on the total number of data tables in a CTPP dataset and the number of tables included in the query result under the label "# of Tables filtered".

Perform a Query

Step 1: Search and select a table

The most efficient way to find a table of interest is first to use filters to narrow down to a set of tables and then use keyword search to find the tables of interest within that set.

Currently, there are four filters for CTPP data tables. They are the Dataset Filter, Part Filter, Universe Filter, and Summary Level Filter. The most important filter of CTPP data tables is the Dataset Filter. This is because the tables are grouped into datasets and each dataset is defined by the period during which the data of the tables are collected. The existing four CTPP datasets were built from American Community Survey (ACS) data collected in year 2000, years 2006–2010, years 2012–2016, and years 2017–2021. Other filters of CTPP data tables, such as the Universe Filter and the Summary Level Filter, may vary from dataset to dataset due to changing focuses of data collection in different period. Therefore, changing the value of the Dataset Filter, i.e., selecting a different dataset, will reset the values of other filters. However, changing the value of other filters will not change the value of the Dataset Filter.

Once the data tables are narrowed down to a small set by applying filters and keyword search, the user can review the meta data associated with each of the resulting tables and decide which table to explore further by retrieving the actual data of that table for review. The user executes their table selection decision by checking the checkbox on the far left of the table's metadata row. The selection of a data table will enable Step 2 of the query for the user to select geography for the query.

Data Table Filters

Dataset

2017-2021 (5-year)

Part

Residence (Part 1)

Universe

All housing units

Summary Levels

(C02) State

(C03) State-C...

(C05) State-PL...

Reset Filters

of Tables filtered:

4 out of 338

Figure 2. CTPP Data Table Filters

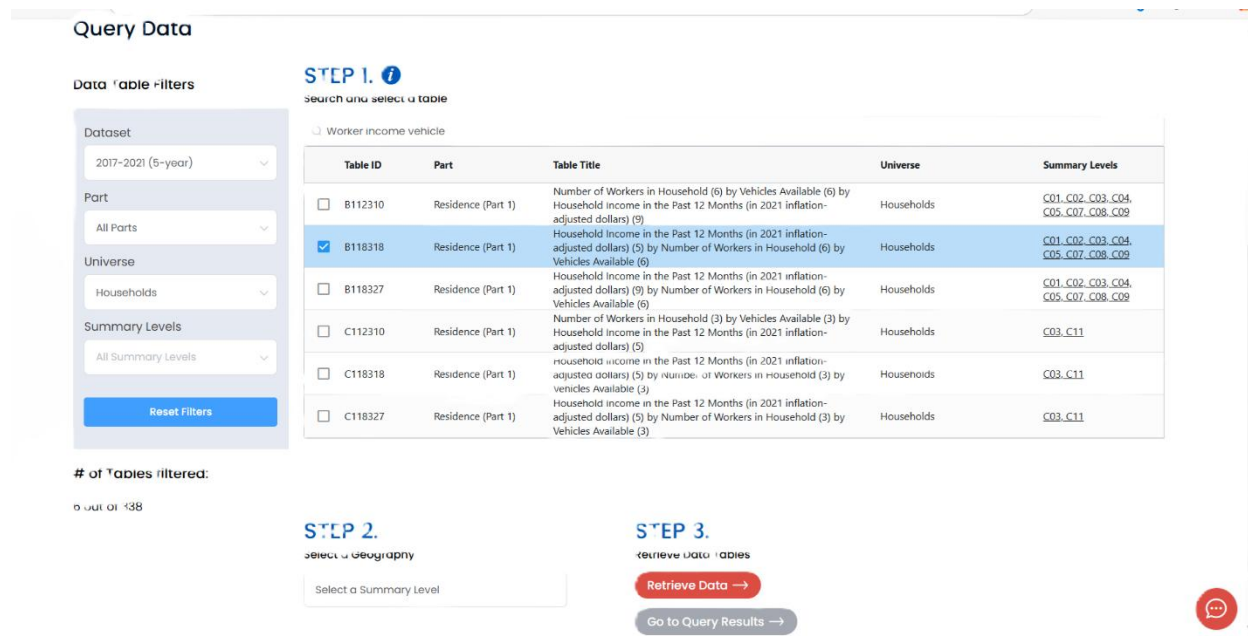


Figure 3. Search and select a table

Step 2: Select Geography

Once a data table is selected, “Step 2: Select a Geography” will be enabled.

In this step of the query, the user must make two types of geography selections:

- Summary level of geography at which the data is aggregated, and
- Specific geographies whose data the user wants to retrieve.

The first geography selection (summary level), made in Step 2-A, determines the spatial unit at which the query data will be aggregated and presented. For example, selecting “State” means the user wants state-level data, while selecting “County” means the user wants county-level data. Figure 4 shows the summary level geographies available for the first geography selection in the CTPP 2012–2016 Flows dataset.

STEP 2.

Select a Geography

State -> State

State-County -> State-County

State-County-MCD (for 12 strong MCD states) -> State-County-MCD (for 12 strong MCD states)

State-Place -> State-Place

Metropolitan Statistical Area - EACH Principal City -> Metropolitan Statistical Area - EACH Principal City

State-PUMA -> State-PUMA

State-Place -> State-County

State-County -> State-Place

State-County-MCD (for 12 strong MCD states) -> State-Place

State-PUMA -> State-Place

Tract -> Tract

TAD -> TAD

TAZ -> TAZ

TAD -> TAZ

TAZ -> TAD

State-Place -> TAZ

TAZ -> State-Place

Figure 4. Query Step 2-A Summary Level Menu

The **second geography selection** (specific geography of interest), made in **Step 2-B**, determines the spatial unit(s) for which data will be returned.

For example:

- Selecting **“Virginia”** and **“Maryland”** indicates that the user wants data for those two states only.
- Selecting **“All States”** indicates that the user wants data for every state.

The user interface for selecting geographies of interest dynamically updates based on two prior user selections:

- The data table selection
- The summary level of geography selection

Impact of Data Table selection

If the user selects a **“Residence” (Part 1)** or **“Workplace” (Part 2)** data table, the available geographies of interest will be **one-dimensional**, meaning the data is associated with a single geographic location.

Example (one-dimensional):

If the user selects a *Residence* data table and chooses **Fairfax County, Virginia**, the data will reflect characteristics of residents who live in Fairfax County only.

If the user selects a **“Flow” (Part 3)** data table, the available geographies of interest will be **two-dimensional**, meaning the data is associated with both:

- an **origin geography**, and
- a **destination geography**.

Since Flow data always represents movement **from residence to workplace**, the **Residence geography is always the origin**, and the **Workplace geography is always the destination**.

Example (two-dimensional):

If the user selects a Flow table and chooses:

- **Residence:** Montgomery County, Maryland
- **Workplace:** Washington, DC

the data will describe people who live in Montgomery County and work in Washington, DC.

Impact of Summary Level of Geography Selection

The **summary level of geography selection** determines the hierarchical levels of geographies available for selection.

For example:

- If **"State"** is selected as the summary level, the interface will display **only a list of states**.
- If **"State–County–Tract"** is selected, the interface will display **three selectable lists**:
 - States
 - Counties
 - Census Tracts

This allows users to drill down the geography selection to the level of detail appropriate to their analysis.

Interface for Selecting Geographies of Interest

Figure 5-1 illustrates the interface for selecting geographies of interest when querying a **Residence** table at the “State–County–Tract” summary level.

The screenshot shows the CTPP Data Portal interface at the URL <https://ctppdata.transportation.org/#/index>. The interface is divided into two main sections: STEP 2 and STEP 3.

STEP 2: Select a Geography

The 'Selected Geographies' section shows 3 items selected. Below it is a table with the following data:

State	County	Tract	Action
Virginia	Fairfax city	Census Tract 3002	
Virginia	Fairfax city	Census Tract 3003	
Virginia	Fairfax city	Census Tract 3004	

The 'Hierarchy' bar shows: State > County > Tract. Below it, instructions state: Check ☒ to select a geography. Click > to expand sub-geographies.

The 'Add to Selection' button is visible. The 'State' panel shows 13 available options, with 'Virginia' selected. The 'County' panel shows 133 available options, with 'Fairfax city, Virginia' selected. The 'Tract' panel shows 5 available options, with 'Census Tract 3002, Fairfax city, Virginia', 'Census Tract 3003, Fairfax city, Virginia', and 'Census Tract 3004, Fairfax city, Virginia' selected.

STEP 3: Retrieve Data Tables

The 'Retrieve Data' button is visible. Below it is a 'Go to Query Results' button.

Figure 5-1. Query Step 2-B Geography of Interest selection
(at the “State–County–Tract” summary Level)

Figure 5-2 illustrates the interface for selecting **origin (Residence)** and **destination (Workplace)** geographies of interest when querying a **Flow** table at the “**State–County**” summary level.

The screenshot displays the CTPP Data Portal interface at the URL <https://ctppdata.transportation.org/#/index>. The interface is titled "STEP 2. Select a Geography" and includes a "Reset Filters" button. Below the title, there is a "Select a Flow Type (defined by user-selected geographies):" section with radio buttons for "Selected to Selected", "Selected to All", and "All to Selected".

The main section is divided into two parts: "Residence" and "Workplace". Each part has a "Selected Geographies" table and a selection interface. The "Residence" section shows "Maryland" selected for the State and "Montgomery County" for the County. The "Workplace" section shows "District of Columbia" selected for both the State and County. The selection interface for each section includes a "State" dropdown menu and a "County" dropdown menu. The "County" dropdown menu is currently expanded, showing a list of counties for the selected state.

At the bottom of the interface, there is a "STEP 3. Retrieve Data Tables" section with buttons for "Retrieve Data" and "Go to Query Results".

Figure 5-2. Query Step 2-B Geography of Interest selection
(for Flow data at the “State-County” summary Level)

There are two key symbols on the geography selection interface that help users navigate and select geographies at each level of the hierarchy of the Summary Level selected in Step 2-A.

- The **checkbox** (☐) allows users to select a geography **and all of its sub-geographies**. This feature is particularly useful when users want to retrieve data for every lower-level geography within a higher-level geography.
- The **greater-than symbol** (>) expands a geography to display its list of sub-geographies, enabling users to drill down and make more granular selections.

Together, these two symbols provide users with a flexible and efficient way to explore geographic hierarchies and select the specific geographies required for their analysis, regardless of the summary level at which the data is requested.

Add Selected Geographies

After selecting a set of geographies of interest, click **“Add to Selection”** to move the selected geographies into the **“Selected Geographies”** section. This step allows users to build a single data query using **multiple sets of geographies across different states**.

Because CTPP data is organized by state, selecting sub-state geographies requires first expanding a state to display its available sub-geographies. These sub-geographies may span multiple hierarchical levels, which can make selection complex. The **“Add to Selection”** function provides a user-friendly way to group and manage sub-state geography selections by state when working with multiple states in a single query.

Review Geography Selections

In the **“Selected Geographies”** section, users can review all selected geographies and remove any that were added unintentionally. Only the geographies listed in this section will be used for data retrieval.

Save Selected Geographies

Geographies listed in the **“Selected Geographies”** section can be saved for future use as **user-defined geographies**. To save a selection, click the **“Save Selection”** button and assign a name to the saved geography set. Saved use-defined geographies can be reused in subsequent queries, eliminating the need to reselect the same geographies.

Copy GeoIDs

The **Geo-IDs** associated with the geographies in the **“Selected Geographies”** section can be copied by clicking **“Copy GeoIDs.”** These Geo-IDs can then be pasted into external programs or scripts that use the **CTPP Data Portal API** to retrieve data programmatically.

Remove Geographies from Selection

Geographies in the **“Selected Geographies”** section can be removed individually by clicking the **trashcan icon** next to a specific geography. To remove all selected geographies at once, click the **“Clear All”** command.

Load Saved Selections

Previously saved user-defined geographies can be loaded into the **“Selected Geographies”** section by clicking **“Load Saved.”** Only saved geography sets that match the **summary level** selected for the current query will be available for loading.

2-Special Queries for Flow Data

Queries of **Flow (Part 3)** data require the selection of both **origin** and **destination** geographies. When a Flow table is selected for a data query, the geography selection interface automatically switches to a mode that enables users to specify **Residence geographies as origins** and **Workplace geographies as destinations**. This default interface is labeled as the **“Selected to Selected”** mode of Flow geography selection, in which user-selected origin geographies are paired with user-selected destination geographies.

In addition to this default mode, the interface provides two **special selection modes** designed to support commonly requested Flow data queries. These modes are labeled **“Selected to All”** and **“All to Selected.”**

- **Selected to All**

This mode allows users to select one or more **Residence geographies** as origins, while **all Workplace geographies** are automatically included as destinations.

- **All to Selected**

This mode allows users to select one or more **Workplace geographies** as destinations, while **all Residence geographies** are automatically included as origins.

These special modes extend traditional **one-to-all** and **all-to-one** querying capabilities by allowing users to define **multiple origins or multiple destinations**, rather than being limited to a single geography.

Select Geographies on Map

“Select on Map” is an alternative method for selecting geographies of interest in the CTPP Data Portal. This option is particularly useful in the following situations:

- **Quick Selections**
It allows users to rapidly select groups of **spatially adjacent geographies**, which may be cumbersome to locate individually in a hierarchical list.
- **Cross-Boundary Selections**
It enables easy selection of geographies that **cross state or county boundaries**, such as **Metropolitan Planning Organizations (MPOs)** or multi-jurisdictional study areas.
- **Low-Level Geography Selections**
It supports selection of **lower-level geographies** (such as Traffic Analysis Zones or Census Tracts) whose names or codes may not be known but can be readily identified based on their **spatial location**.

To select geographies using the **“Select on Map”** method, click the **“Select on Map”** link on the geography selection interface. An interactive map will open, providing several selection tools, including:

- Click to select individual geographies
- Draw a polygon to select multiple geographies
- Draw a circle to select geographies within a specified area

All geographies selected on the map are automatically added to the **“Selected Geographies”** section and are immediately available for data retrieval. Once all desired selections have been made, click **“Back to Query”** to return to the main query interface.

Note: The “Select on map” function is very useful for several key reasons:

- 1) It enables quick selections of a group of geographies adjacent to each other spatially but far away from each other in the hierarchical list.
- 2) It enables easy selection of geographies within a region that cross State or County boundaries, such as an MPO.
- 3) It enables easy selections of geographies at low spatial levels, such as TAZ or Census Tract, whose name or code is not known to the user, but its spatial location is.

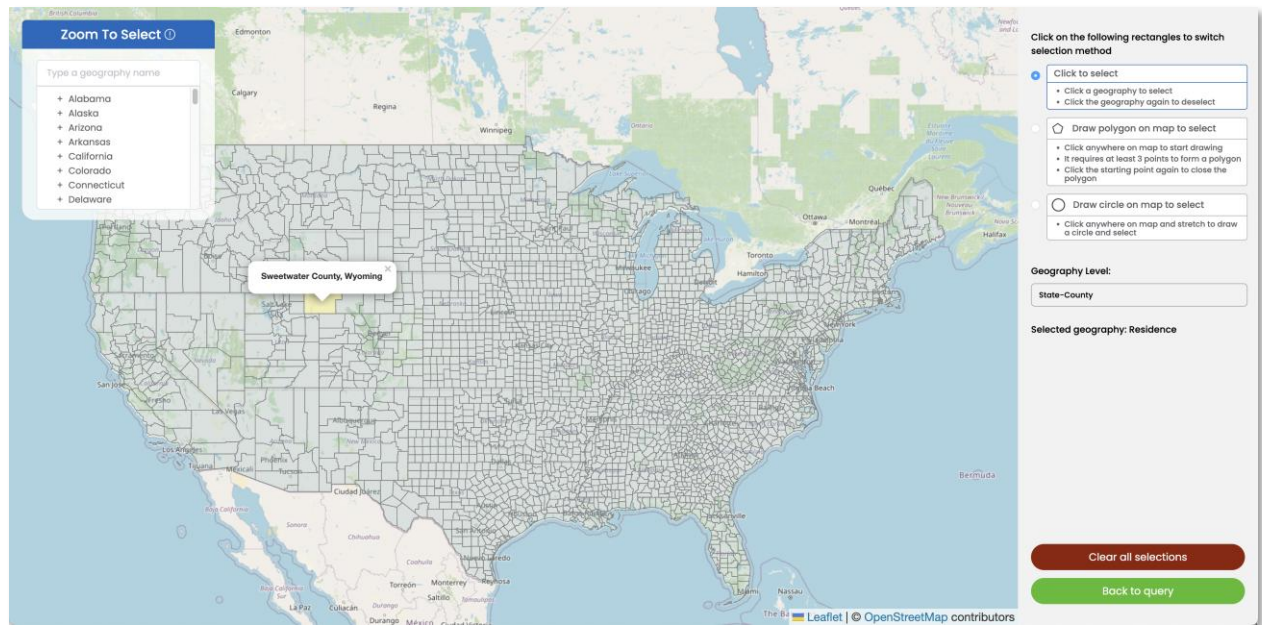


Figure 6. Select on Map Interface for Selecting Geographics of Interest

Table 2 presents key features of the map selection interface.

Table 2. Select on Map Option Key Features

Feature	Utilities of the feature
Mouseover	This feature displays the name of the geography over which the mouse pointer is hovering.
Zoom	This feature allows the user to zoom in and zoom out with the map by turning the wheel of the mouse.
Click to select	This option allows a user to select a geography on the map by clicking it and deselect it by clicking it again. "Click to select" is the default mode of selection. This means two things: 1) The mode of selection will be "Click to select" unless the user specifies to use other mode of selection. 2) The mode of selection will default back to "Click to select" every time a selection is completed no matter which mode of selection was used to perform the selection.
Draw polygon to select	This mode of selection enables all geographies to be selected on the map touched by the polygon drawn by the user. To draw a polygon, the user needs to select (by clicking) at least 3 points on the map and click the starting point again to close the polygon. Once a polygon selection is completed, the mode of selection will automatically be returned to the default mode "Click to select." To perform another polygon selection, the user must select the "Draw polygon to select" again. The reason for this design is to make it easy for the user to use "Click to select" to further refine the selections of geographies resulting from the "Draw polygon to select", which may miss some wanted geographies but include some unwanted geographies.

<i>Draw circle to select</i>	This mode of selection selects all geographies on the map touched by the circle drawn by the user. To draw a circle, the user can click anywhere on the map, hold the click, and stretch the mouse away from the center until the desired radius is reached. Once a circle selection is completed, the mode of selection will automatically be returned to the default mode "Click to select." To perform another circle selection, the user needs to select the "Draw circle to select" again. The reason for this design is to make it easy for the user to use "Click to select" to further refine the selections of geographies resulting from the "Draw circle to select", which may miss some wanted geographies but include some unwanted geographies.
<i>View Geography level</i>	This feature shows the summary level geography specified by the user for the query. To change the geography level, the user must exit the map selection and return to the main query selections.
<i>View selected geography</i>	This feature shows (1) what type of geography (residence, workplace, or origin and destination of flows) is being selected and (2) the names of the geographies that have been selected. The names of the geographies selected is dynamically generated, showing the latest result of the user's selections and deselections. The Selected geographies include all geographies selected no matter if the selections are made via Map Selection or List Selection.
<i>Zoom to select</i>	This feature allows the map to automatically zoom-in to a geography the user selected from the hierarchical list of geographies and center it on the map, so that the user can select on the map the geographies in the vicinity of the zoomed-in geography. This feature is particularly helpful to the user in selecting small geographies such as Census Tract and Transportation Analysis Zone (TAZ), which would not be recognizable without zooming-in.
<i>Clear all selections</i>	This feature clears all selections made on the map with one click. So, the user can restart making selections.
<i>Back to query</i>	This feature allows the user to go back to the query specification section with one click. This feature is particularly useful when the user to go back and forth between "Geography list selection" and "Geography map selection".

Note: The summary level and geographies of interest options may differ from one CTPP dataset to another. The summary level geographies can differ between Parts within the same dataset year. For example, the summary level geographies for Year 2000 Flows dataset are different from those of Year 2000 Residence dataset (because the summary level geographies of the Flows dataset can show origin and destination at different levels). Also, the summary level geographies of the Year 2000 Workplace dataset are different from those of the Year 2000 Residence dataset (e.g., the "State-Public Use Microdata Sample Area (PUMA)" and the "Urbanized Area (UA)"). There are even more differences in summary level geographies and geographies of interest among CTPP datasets across CTPP data years, especially between Year 2006-2010 and Year 2000. The portal dynamically determines the summary level geography and geographies of interest selection options based on the dataset and Part selected to aid users in navigating the CTPP data even given these discrepancies.

Step 3: Retrieve Data

The “Retrieve Data” button will be enabled when the required selections in Step2 are made. Selecting the button will execute the query and present the query results (Figure 8) directly below the query selection options.

STEP 3.

Retrieve Data Tables

Retrieve Data →

Go to Query Results →

Figure 7. Query Step 3
Retrieve Data Button

Query Results

Results from your query are displayed below. Use the left Data Menu to change what data is visible within each table.

1 Tables Returned Under This Query						
Household Income in the Past 12 Months (in 2021 inflation-adjusted dollars) (9) by Number of Workers in Household (3) t v						
Summary Statistics	Measures	Table Info	Transpose	Blank Rows	Chart	Download
Share						
Household Income in the Past 12 Months (in 2021 inflation-adjusted dollars) (9)	Total, household income					
Number of Workers in Household (6)	Total, number of workers in household					
Vehicles Available (6)	Total, vehicles available		No vehicle available		1 vehicle available	
RESIDENCE v	Estimate v	Margin of Error v	Estimate v	Margin of Error v	Estimate v	Margin of Error v
Alabama	1,902,985	5,848	106,650	2,514	600,210	1,320
Alaska	260,560	1,333	23,870	811	80,320	1,320
Arizona	2,683,555	6,008	149,800	3,449	926,260	1,320
Arkansas	1,158,460	3,726	68,775	1,890	377,650	1,320
California	13,217,585	19,813	911,655	7,833	3,960,945	1,320
Colorado	2,227,930	3,827	110,960	2,593	658,025	1,320
Connecticut	1,397,325	3,453	118,175	2,569	460,845	1,320
Delaware	381,095	1,554	22,370	1,364	130,475	1,320
District of Columbia	310,105	1,721	110,825	2,566	140,080	1,320
Florida	8,157,420	21,975	489,115	6,674	3,165,110	1,320
Georgia	3,885,370	8,943	236,210	4,529	1,251,765	1,320
Hawaii	478,415	2,006	37,990	1,489	157,200	1,320
Idaho	657,100	2,561	25,085	1,324	165,280	1,320
Illinois	4,930,255	7,843	526,290	5,772	1,713,320	1,320

Figure 8. Sample Query Results

3. Query Result Display

The query result display contains four main sections (see screenshot below):

- 1) Title of data table returned
- 2) Data table
- 3) Data menu
- 4) Data utilities

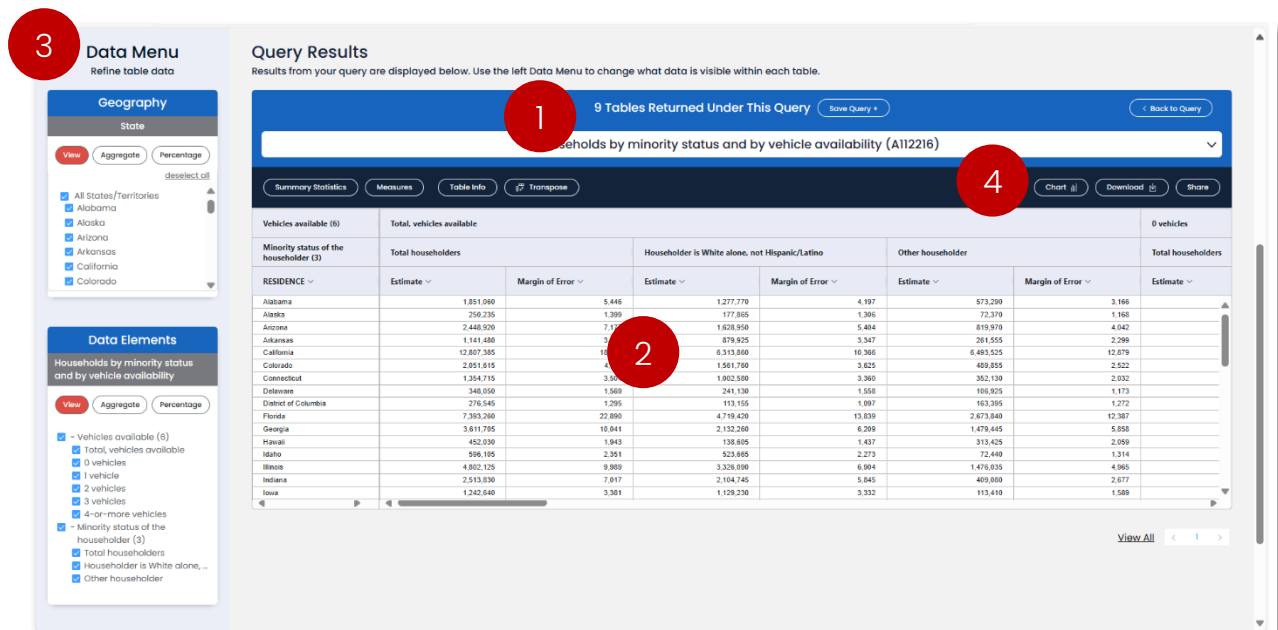


Figure 9. Query Result Display with Four Main Sections

Title of data table returned

The title of the data table returned by the query is provided here. The title provides brief descriptions of what kind of data is contained in the returned data table.

Data Table

The results of a data query are always presented in table format, with columns representing the data elements and rows representing the geographies of interest. Table presents key features of the data table.

Table 3. Data Table Features

Feature	Utilities of the feature
Column header	Column headers of a table provide the following information about the data in the table: Unit of measure of the data, such as number of vehicle, people, worker, or household, which the primary data of the table stand for. Dimensions of the data, such as age group, income level, employment status, or race by which the data are aggregated into columns of the table. Statistical measure of the data, such as estimate, marginal of error (MOE), standard error, etc. in which the data are presented.
Row header	Row headers of a table show the geographies of data table in the format: "Name of geography of interest, Name of geography of higher level". The number of higher levels can be 2, 1, or 0, depending on what is the geography of interest. For example, if states are the geography of interest, then the number of higher levels will be zero; if counties are the geography of interest, then the number of higher levels will be one, i.e., the state; if Census Tracts are the geography of interest, then the number of higher levels will be two, i.e., County, State.
Vertical scroll	Vertical scroll of the data table section allows the user to scroll a data page up and down to see all the data rows on a page.
Horizontal scroll	Horizontal scroll of the data table section allows the user to scroll a data page left and right to see all the data element columns of a data table.
Pagination numbers	When the query result table contains a significantly large number of rows, the table will be divided into multiple pages along the row dimension to improve user experience and performance. Pagination numbers are displayed at the bottom right corner of the Data table section.

Data Menu

The Data Menu offers users four options to customize the queried data:

- Create a customized view of the Query Results.
- Create aggregates of geographies or data categories.
- Calculate percentages of selected geographies or data elements over a specified denominator geography or data element, respectively.
- Show data in time-series format for comparisons across years.

Create a Customized View of the Query Results

The default of query result display is to present all contents in a displaying data table for the user to view. This means that all data elements and all geographies of that table are selected for displaying. For example, in Figure 10, all geographies and all data elements are selected for displaying by default.

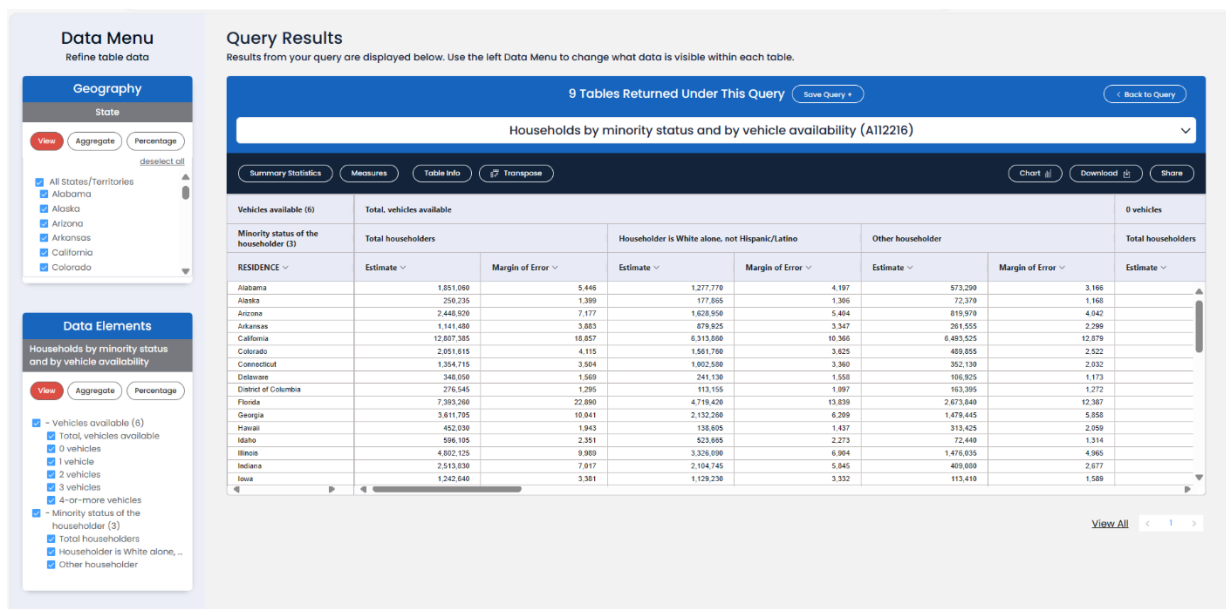


Figure 10. Default Display of Query Results (All Queried Geographies and Data Elements)

To create a customized view of a table, the user selects only a subset of geographies or data elements or both from the Data Menu to be displayed in a revised data table. For example, as shown in Figure 11, only three geographies (Alabama, Arizona, and California) and two data elements (0 vehicles, Householder is White alone..., other householder) are selected in a customized view.

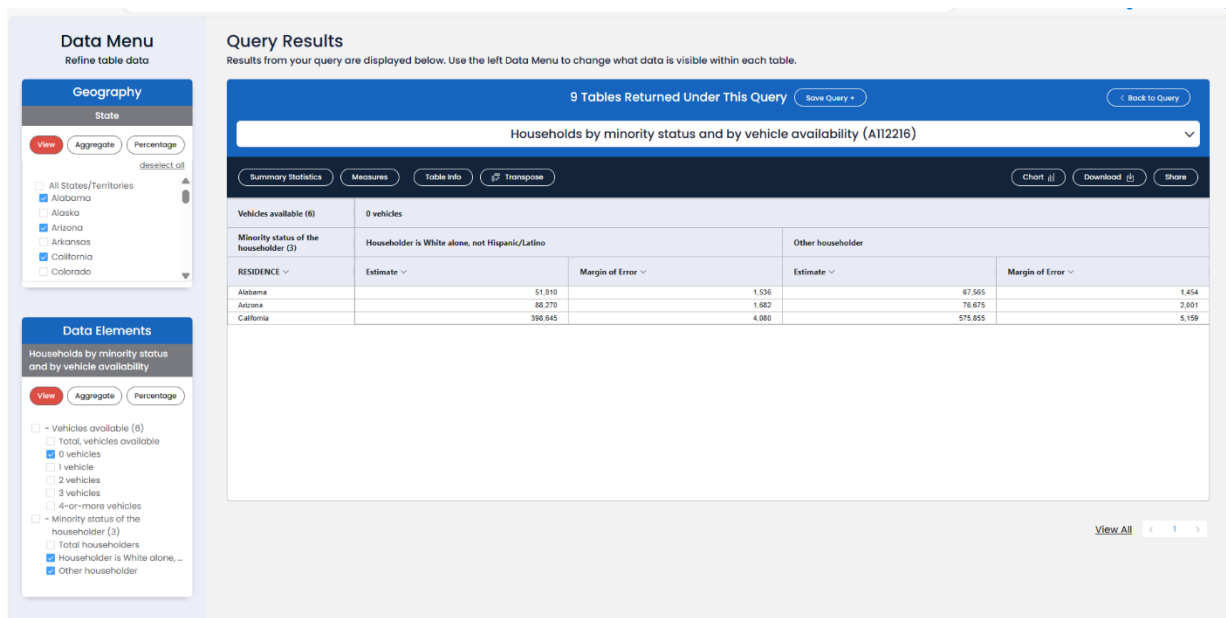


Figure 11. Customized View of the Data Table

Note: Users should be very careful when selecting data elements to customize the view of data in a data table. Data elements in CTPP data tables are usually defined by multiple dimensions, such as number of households by age by income and by race. This means multiple selections (at least one selection along each dimension) are needed to specify a particular data element (or column) in the data table. Also, with each dimension there is always a “Total” category. For example, the vehicle availability dimension may have the following categories: total vehicles available, 0 vehicle, 1 vehicle, 2 vehicles, 3 vehicles, and 3+ vehicles. When there is no selection made for a data element dimension, the “total” category will be selected as default to display data. That is, no selection for a data dimension is equivalent to selecting the “total” category for that dimension. Figure 12 shows that no selection was made for either of the two data element dimensions of the table, Vehicle availability and Minority status of households, but the “Total, vehicles available” and the “Total households” were shown in the data display by default.

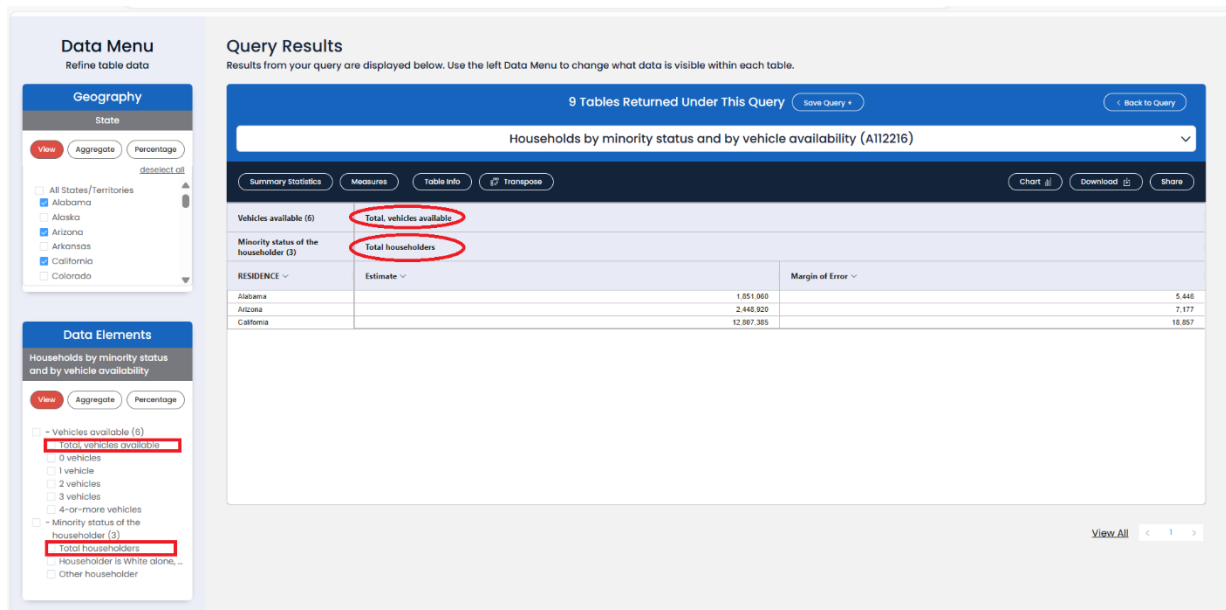


Figure 12. Sample Custom Data View with No Selection for Some Data Element Dimensions

Create Aggregate Geographies

The user completes four basic steps to aggregate geographies.

- Step 1: Select the “Aggregate” button in the Geography section of the Data Menu to initiate the aggregate function.
- Step 2: Select the “Add aggregate” label to pop up the interface for selecting geographies to be aggregated.
- Step 3: Select the geographies to be aggregated and enter a name for the resulting aggregate geography.
- Step 4: Select the “Calculate” button to execute the aggregation.

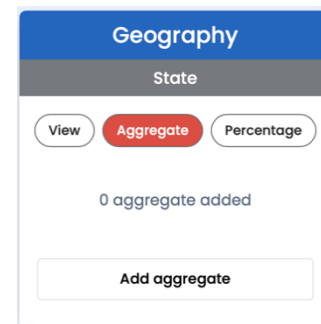


Figure 13. Aggregate Geographics Option

The result of the aggregation is shown in Figure 14.

- A data row for the new Aggregate geography, 4-States, was added to the top of the data table and highlighted in yellow.
- The name of the new Aggregate geography, 4-States, is listed under the “Aggregate” button, where the name can be edited or deleted.

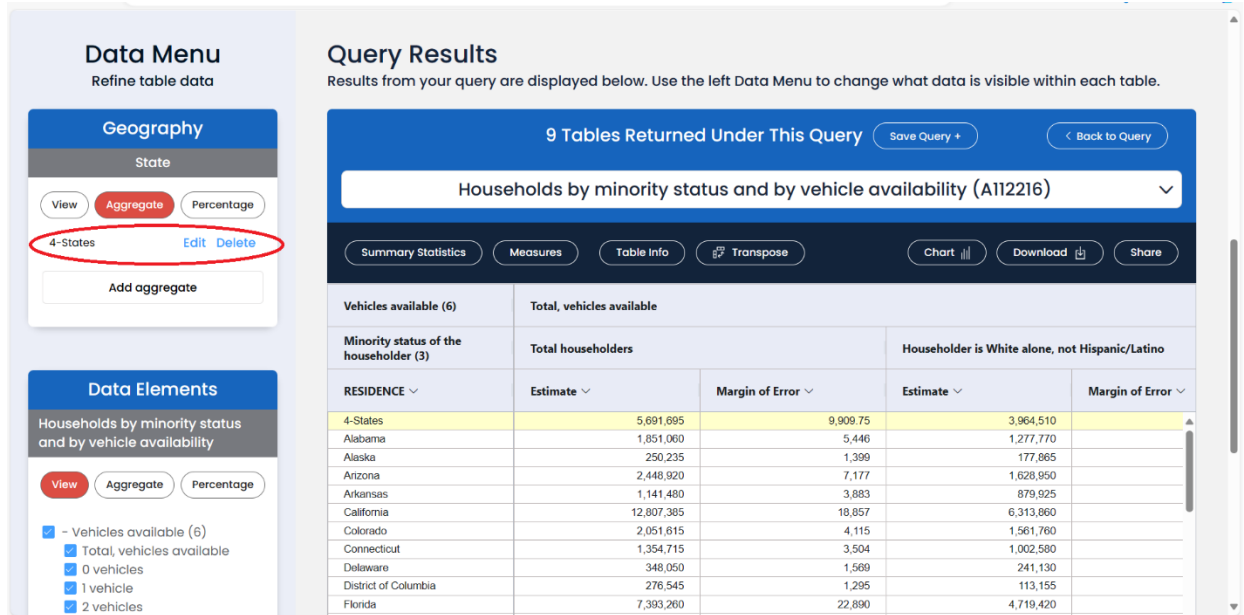


Figure 14. New Aggregate Geography Added

To create another Aggregate geography, the user simply repeats the process.

Create Aggregate Data Elements

The process to create aggregate data categories is like that for creating aggregate geographies, and requires four basic steps:

- Step 1: Select the “Aggregate” button in the Data Elements section of the Date Menu to initiate the aggregate function.
- Step 2: Select the “Add aggregate” button to view the interface for selecting data categories to be aggregated.
- Step 3: Select the data elements to be aggregated and enter a name for the resulting aggregate data category.
- Step 4: Select the “Calculate” button to execute the aggregation.

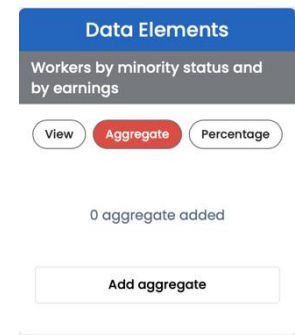


Figure 15. Aggregate Data Element Option

The result of the aggregation is shown in Figure 16.

- Multiple data columns for the new Aggregate category, Multiple Vehicles, were added to the far left of the data table and highlighted in yellow.
- The name of the new Aggregate category, Multiple Vehicles, is listed under the “Aggregate” button, where the name can be edited or deleted.

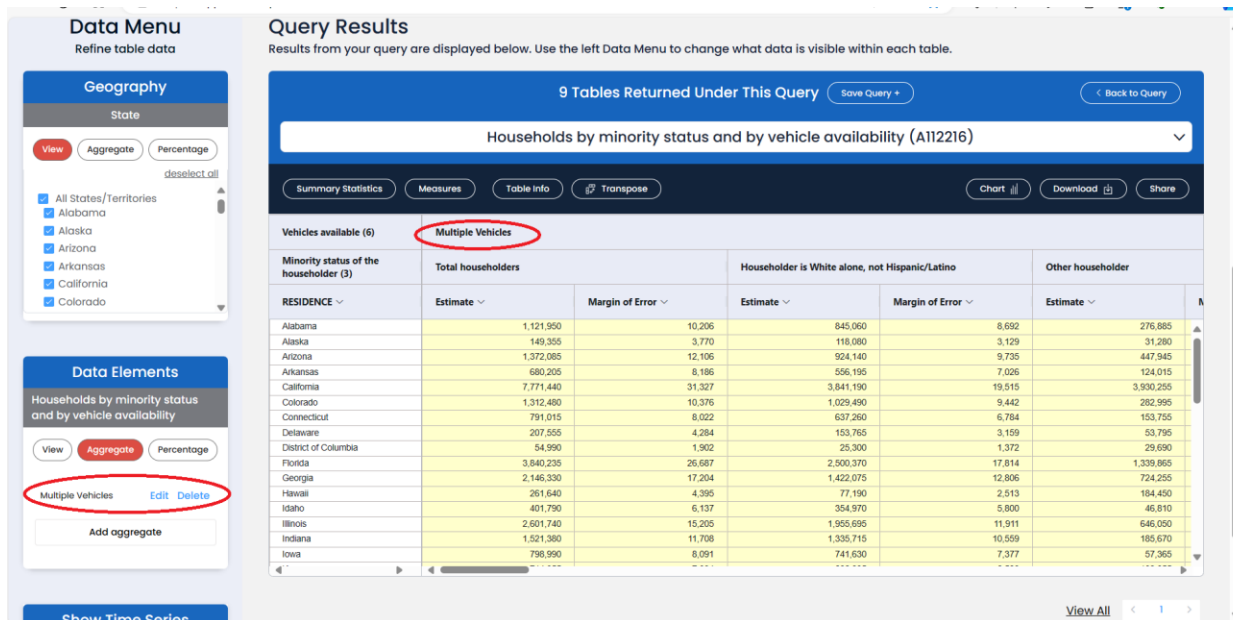


Figure 16. New Aggregate Data Element Added

To create another Aggregate data element, the user simply repeats the process.

*Note: CTPP data elements usually come with multiple dimensions that define them, such as number of households by vehicle availability by race. Aggregate data categories can only be created by aggregating data elements along the same dimension. For example, a "Multiple Vehicles" category can be created by aggregating the 2 vehicles, 3 vehicles, and 4 vehicles data elements in the vehicle availability dimension. Likely, a multi-races category can be created by aggregating Black and Asian categories in the race dimension. Aggregation across multiple data dimensions **cannot** be calculated. Consequently, once a selection is made in one data dimension, other data dimensions will not be available to select.*

Calculate Percentages by Geography

The process of calculating percentages by geography includes the following steps:

- Step 0: Select the Percentage button in the Geography section of the Data Menu to bring up the interface for percentage calculations.

Note: "Step 0:" is used here so that the following step numbers are consistent with those shown in the interface for percentage calculations.

- Step 1: Select one or multiple geographies to be used as numerators from the dropdown list of the "Select Numerators" field. Multiple geographies can be selected. User created Aggregate geographies are also available to be used as numerators.
- Step 2: Select a geography to be used as denominator from the dropdown list of the "Select Denominator" field. Only one geography can be selected. User created Aggregate geographies are also available to be used as denominators.

- Step 3: Select the “Calculate” button to execute the percentage calculation.

Figure 17 provides an example of percentage calculation by geography, where:

- State Alabama, Alaska, Arizona, Arkansas, California, and the user-created aggregate geography 4-States are used as numerators.
- The aggregate geography, 4-States, is used as the denominator.
- The calculation results are presented in the data table display section, with the user created aggregated geography highlighted in yellow.

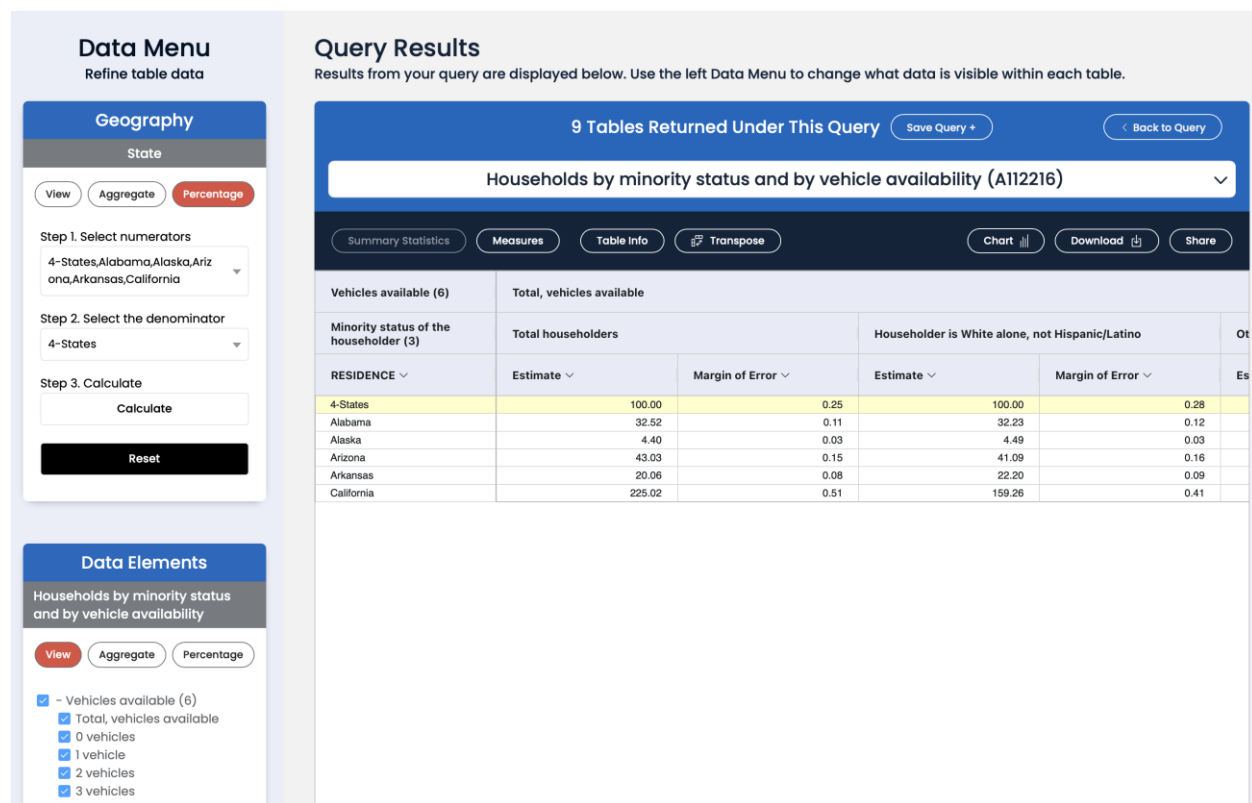


Figure 17. Sample Percentage Calculation by Geography

Calculate Percentages by Data Element

The process of calculating percentages by data element includes the following steps:

- Step 0: Select the Percentage button in the Data Elements section of the Data Menu to bring up the interface for percentage calculations.

Note: “Step 0:” is used here so that the following step numbers are consistent with those shown in the interface for percentage calculations.

- Step 1: Select one or multiple data elements to be used as numerators from the dropdown list of the “Select Numerators” field. Multiple data elements can be

selected. User created Aggregate data elements are also available to be used as numerators.

- Step 2: Select a data element to be used as denominator from the dropdown list of the “Select Denominator” field. Only one data element can be selected. User created Aggregate data elements are also available to be used as denominators.
- Step 3: Click the “Calculate” button to execute the percentage calculation.

Note: Since a data element (column) in a CTPP data table may be defined by multiple dimensions, each of those dimensions must be selected to specify a specific data element in selecting a denominator or a numerator.

Figure 18 provides an example of percentage calculation by data element, where:

- The data elements (columns): Total, vehicle available/Total Householders, 0 vehicle/Total Householders, 1 vehicle/Total Householders, 2 vehicles/Total Householders, 3 vehicles/Total Householders, 4 vehicles/Total Householders, are used as numerators.
- The “Total, vehicle available/Total Householders” data element (column) is used as the denominator.

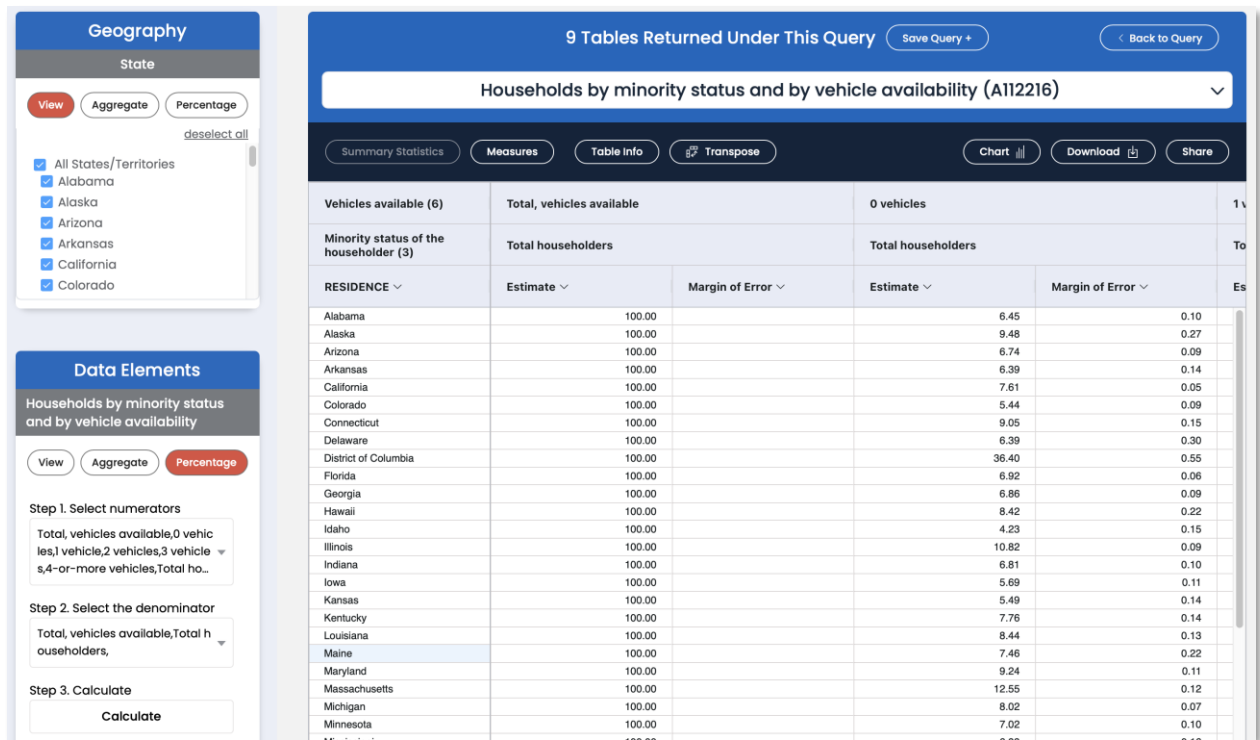


Figure 18. Sample Percentage Calculation by Data Element

Show Data in Time Series Format for Comparison Across Years

The “Show Time Series” feature at the bottom of the Data Menu section allows users to:

- Find out if the data generated by a query is available in other CTPP data years.
- Put other years’ data, if available, side by side to the data generated by a query to form a time series.
- Organize by the time series data by variable or by year. The time series by variable format groups the same data element from different years together to enable comparison analysis across years. The time series by year format groups data of each year together and concatenates the yearly datasets together.

Figure 19 shows time series data by variable, while Figure 20 shows time series data by year.

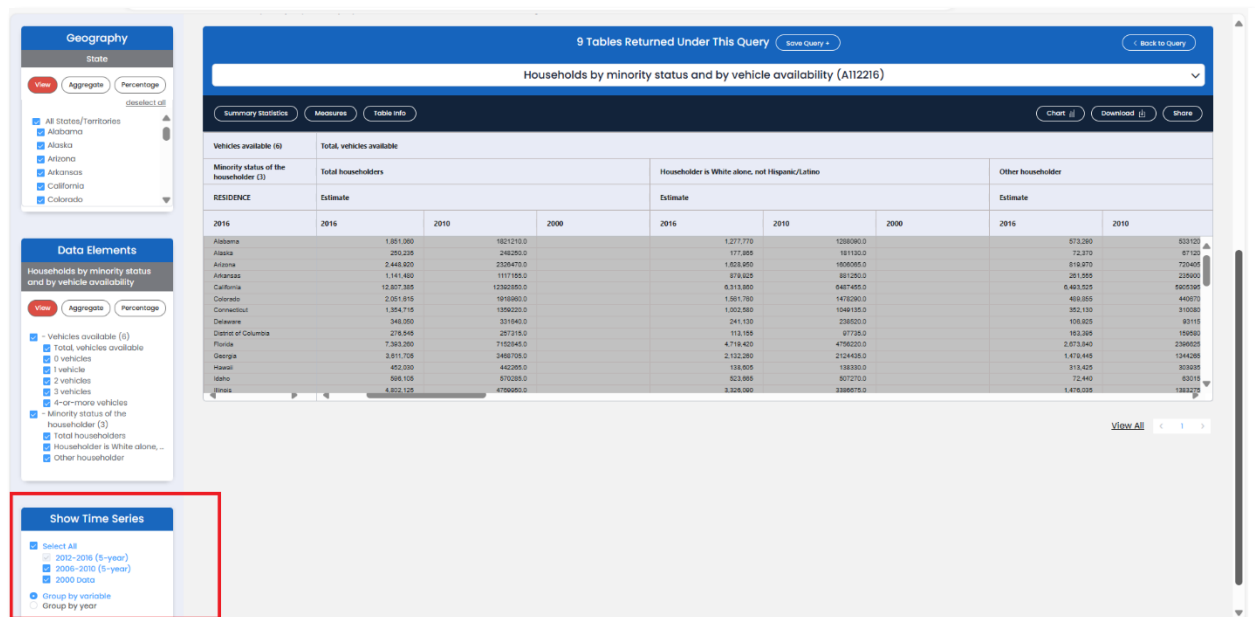


Figure 19. Sample Time Series Data by Variable

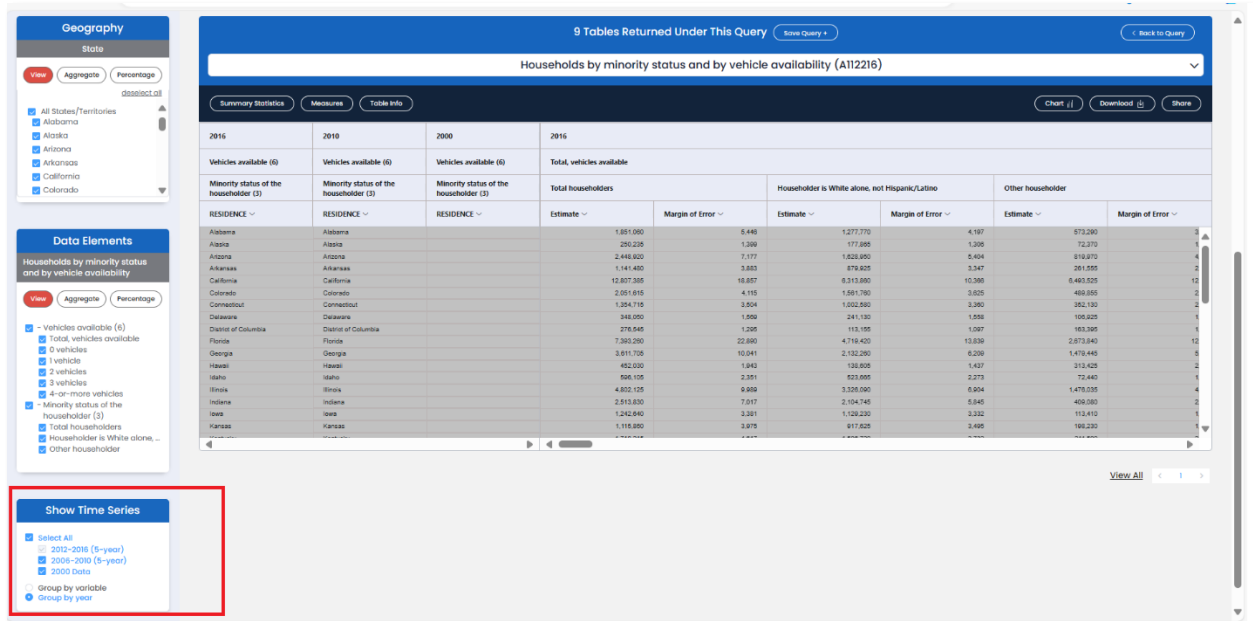


Figure 20. Sample Time Series Data by Year

Data Utilities

Table 4 presents key features of the data utilities section.

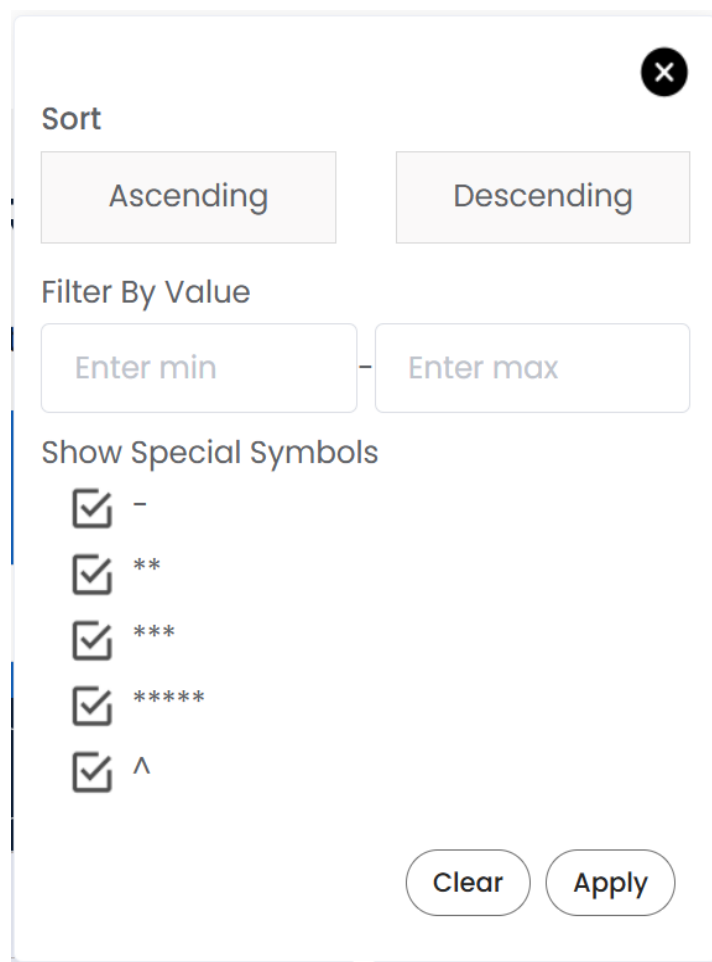
Table 4. Data Utilities Features

Feature	Utilities of the feature
Summary Statistics	The "Summary Statistics" button provides a list of summary statistics that can be derived from and added to the displayed data table. If the user selects summary statistics from the list, it will be added to the top of the data table.
Measures	The "Measures" button provides a list of statistical measures that can be calculated for and added to the displayed data table. The selected statistical measures will be calculated for and added to all data elements in the data table.
Table Info	The "Table Info" button provides the metadata of the displayed data table, such as Table ID, Table contents, Universe of the table data, etc. It also include explanations of special symbols in CTPP data tables.
Transpose	The "Transpose" button converts rows of the displayed data table into columns and vice versa. As a result, the rows of the original table become the columns of the transposed table, and the columns become the rows.
Blank Rows	By default, blank rows of query result will be removed and therefore will not show in the query result table display. However, users can include blank rows in the table display and the data download file by selecting the "Blank Rows" button. Clicking the "Blank Rows" button again will remove the blank rows from the query result table display and the data download file.
Sort	Query result table can be sorted ascending or descending by the values of any of the columns of the data table. Clicking on the "▼" symbol next to the column header to bring up the interface of the sorting function. See more details in Sorting Query Result section below.

Chart	The “Chart” button takes the user to the Data Visualization function, which is described in the following section, Query Data Visualization .
Download	The “Download” button is at the center-right of the query result display section. This function enables the user to download a copy of the query result to a storage space specified by the user.
Share	The “Share” button is at the center-right of the query result display section. This function allows the user to generate a link to the query result and share it with others by email, text message, web posting, etc.

Sorting and Filtering Query Result

CTPP data query result can be sorted ascending or descending or filtered by values between a range specified by the user. These sorting abilities allow users to easily identify the top or the largest rows and the bottom or the smallest rows based on the values of the data column used for sorting. The sorting also allows the user to identify rows whose values fall within a range specified by the user. The user can initiate the sorting function by clicking the “” symbol next to the header of a data column. See Figure 21.



×

Sort

Ascending

Descending

Filter By Value

Enter min

-

Enter max

Show Special Symbols

☒ -
☒ **
☒ ***
☒ *****
☒ ^

Clear

Apply

Figure 21. Query Result Sorting

- Sort ascending: Selecting the Ascending button will cause the rows of the query result to be sorted ascending from the smallest value to the largest values of the sorting column.
- Sort descending: Selecting the Descending button will cause the rows of the query result to be sorted descending from the largest value to the smallest value of the sorting column.
- Filter by value: Specifying the minimum and the maximum values will cause the query result to show only those rows whose values of the sorting column fall between the minimum and maximum values (inclusive).
- Show special symbols: When a special symbol is selected, the system will assign a numeric value to it. Consequently, sorting and filtering will evaluate that numeric value and place the rows with that special character accordingly in the sorted or filtered data set.

Explanations of Special Data Symbols

CTPP data tables contain some special symbols. The meanings of these special symbols are presented below.

– The estimate could not be computed because there was an insufficient number of sample observations. For a ratio of medians estimate, one or both of the median estimates falls in the lowest interval or highest interval of an open-ended distribution.

N The estimate or margin of error cannot be displayed because there were an insufficient number of sample cases in the selected geographic area.

(X) The estimate or margin of error is not applicable or not available.

median– The median falls in the lowest interval of an open-ended distribution (for example "2,500–")

median+ The median falls in the highest interval of an open-ended distribution (for example "250,000+").

** The margin of error could not be computed because there were an insufficient number of sample observations.

*** The margin of error could not be computed because the median falls in the lowest interval or highest interval of an open-ended distribution.

***** A margin of error is not appropriate because the corresponding estimate is controlled to an independent population or housing estimate. Effectively, the corresponding estimate has no sampling error and the margin of error may be treated as zero.

4. Query Data Visualization

Access the Data Visualization Interface

The CTPP data portal has a visualization interface for users to define the data to be used to generate the visualization and the type of visual they want to create. The user accesses the visualization interface by selecting the “Chart” button in the query result display section, as shown in Figure 21.

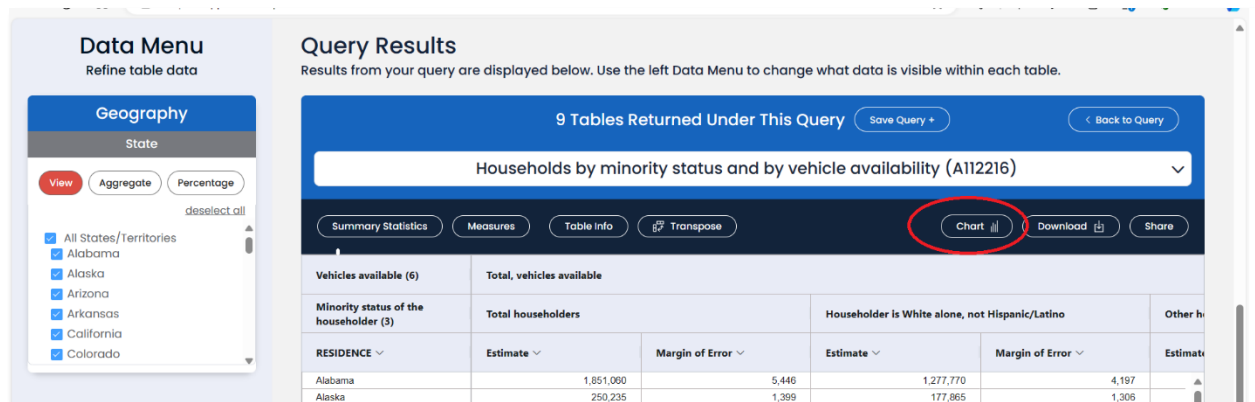


Figure 21. Chart Button Used to Access Data Visualization Interface

Note: Since the query result display section only appears when there is a query result to display, the “Chart” button will only appear and therefore the visualization interface will only be accessible when there is a query result.

Creating a Visualization in Four Steps

Upon selecting the “Chart” button, the Data Visualization interface will open up with the data table returned from the query readily available for visualization and the four Steps for creating a visualization clearly displayed (Figure 22). The user does not need to take any action to import the data table.

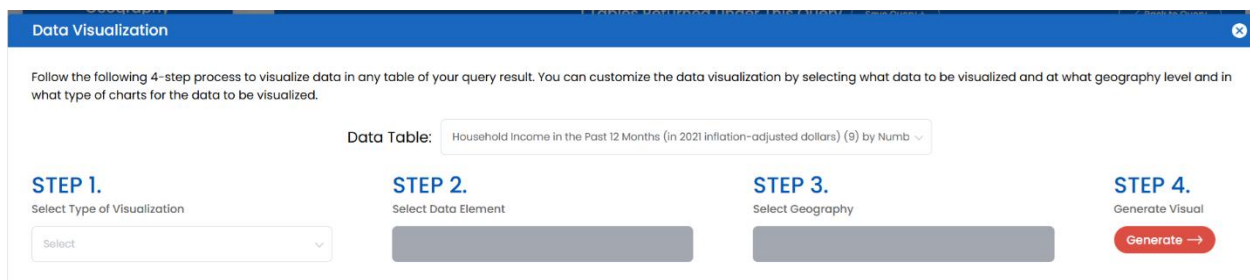


Figure 22. Four Steps for Creating a Visualization

Select Type of Visual and Data to Create the Visual

To create a data visualization, the user completes four steps:

- Step 1: Select Type of Visualization
- Step 2: Select Data Elements
- Step 3: Select Geography
- Step 4: Generate Visual

Step 1: Select Type of Visualization

The user can choose from eight chart types for visualization, as shown in Figure 23:

- Bar chart
- Column chart
- Column chart map
- Line chart
- Pie chart
- Thematic map
- Spider map
- Sankey diagram

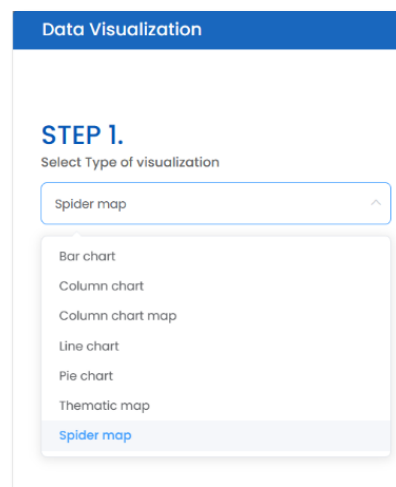


Figure 23. Visualization Type Dropdown

The user can only select one visual type at a time from the dropdown list. If “Pie chart” is selected, the user must specify visualization “By data element” or “By geography”.

Note: Not all chart types are supported by all data tables, and, therefore, some chart types may not be available for certain data tables. For example, the spider map depicts the flow for a topic of interest from a residence to a workplace, and, therefore it is only available for Flow data tables. If the data table for visualization is a Residence table or a Workplace table, then the Spider Map will not be available.

Step 2: Select Data Elements

The user selects the checkbox next to the Data Elements to be included in the visual as shown in Figure 24.

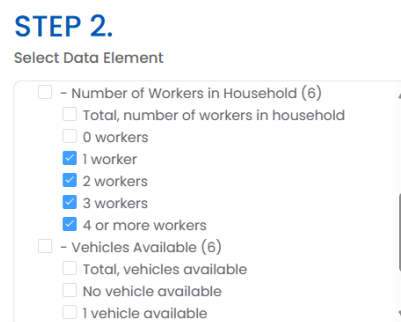


Figure 24. Visualization Data Element Menu

Note: The Data Elements available for visualization are determined by the data table to be visualized. They are also limited by the visual type selected in Step 1. For example, thematic map and pie chart by location will allow the selection of only one topic of interest.

Step 3: Select Geographies

The user selects the checkbox next to the geographies to be included in the visual.

Only one geography or one pair of geographies can be selected if the visualization type is a pie chart by variable. For all other visualization types, all geographies or pairs of geographies specified in the query result are available for selection.

Step 4: Generate Visual

Click the “Generate” button to create the specified visual with the selected data.

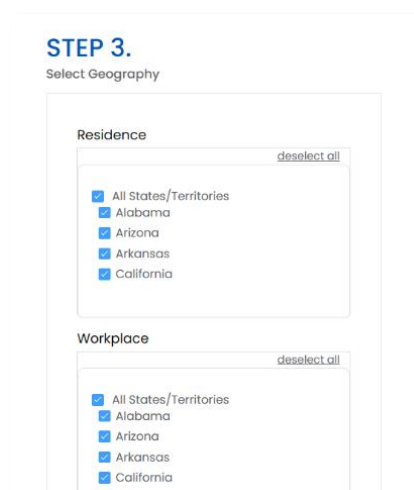


Figure 25. Visualization Geography Menu

Change Visualization Type

The user can change the visualization type after creating a visual by selecting the “Switch chart type” option above the visual, as shown in Figure 26. The visualization will change immediately upon selection of a different visual type from the dropdown.

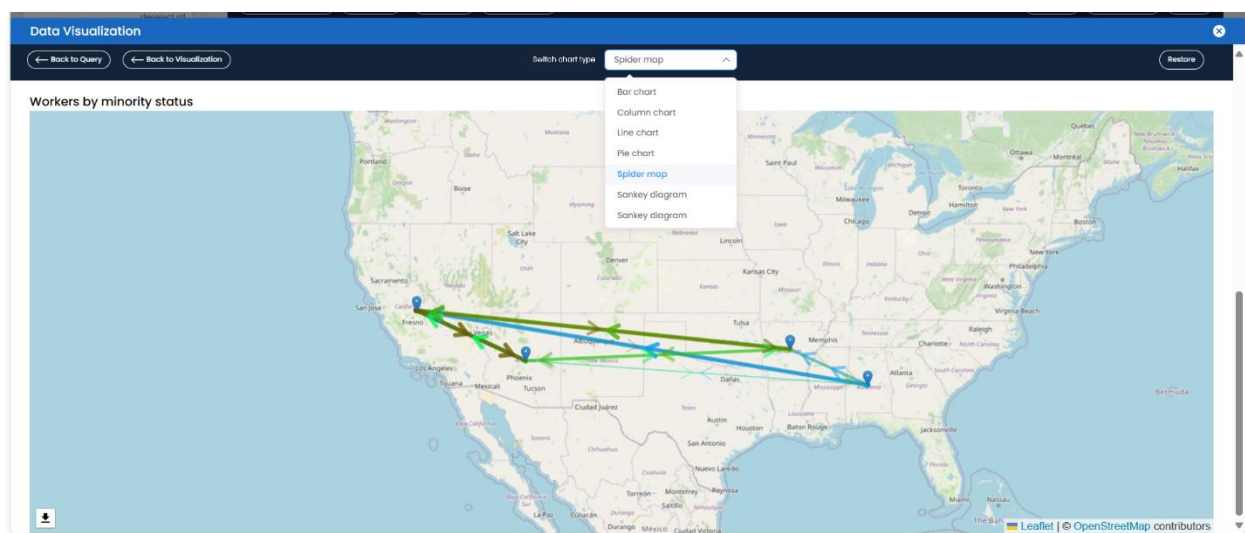


Figure 26. Switch Chart Type Menu

Note: The dropdown list is dynamically determined based on the dataset, topic(s) of interest, and geography selections for the visualization. The list only includes chart types that are suitable for the visualization specifications.

5. General Utilities

The CTPP Data Portal offers general utilities to aid users in working with CTPP data. Table 5 presents these general utilities.

Table 5. General Utilities Available in the CTPP Data Portal

<i>Utility</i>	What, why, and how of the Utilities
<i>Create account</i>	Users can create a CTPP Data Portal account to save queries, reuse and manage saved queries, and generate and manage API keys. To create an account, the user completes the following steps: • Step 1: Select the “Login” button at the top right corner of the page. • Step 2: Select the “Create an account” link below the Login button. • Step 3: Complete all fields in the “Create An Account” window. • Step 4: Select the “Create an Account” button.
<i>Login</i>	Users with a CTPP Data Portal account can login to their account by selecting the “Login” button in the top right corner of the portal.
<i>Reset password</i>	The user can reset the account password by completing the following steps: • Step 1: Select the “Login” button at the top right corner of the page. • Step 2: Select the “Forgot password” link. • Step 3: Insert the email address associated with the portal account. • Step 4: Select the “Submit” button.
<i>Save query</i>	The user with a portal account can save a query after it is generated. To save a query first generates a query, then selects the “Save Query” button located on the top of the query result display section. The user must be logged in to save a query.
<i>Link to API Explorer</i>	Users can access the CTPP Data Application Programming Interface (API) by selecting the “API Explorer” link in the top right corner of the portal. The CTPP Data API is a data service that enables software developers to access and use CTPP data within their applications.
<i>Link to Data Portal home from API</i>	Within the CTPP Data API, users can select the “Data Portal” link in the top right corner of the API to return to the CTPP Data Portal.

6. API Explorer

The CTPP Data API (or API Explorer) is a powerful tool for visualizing and interacting with APIs described using the OpenAPI Specification (OAS), formerly known as Swagger Specification. This guide will help you navigate through the API Explorer and make the most out of its features.

Generate an API Key

The CTPP Data API requires a valid API key. To get an API key, users can follow the process described as follows.

- Go to the CTPP Data Portal and login. (Register if no account exists.)

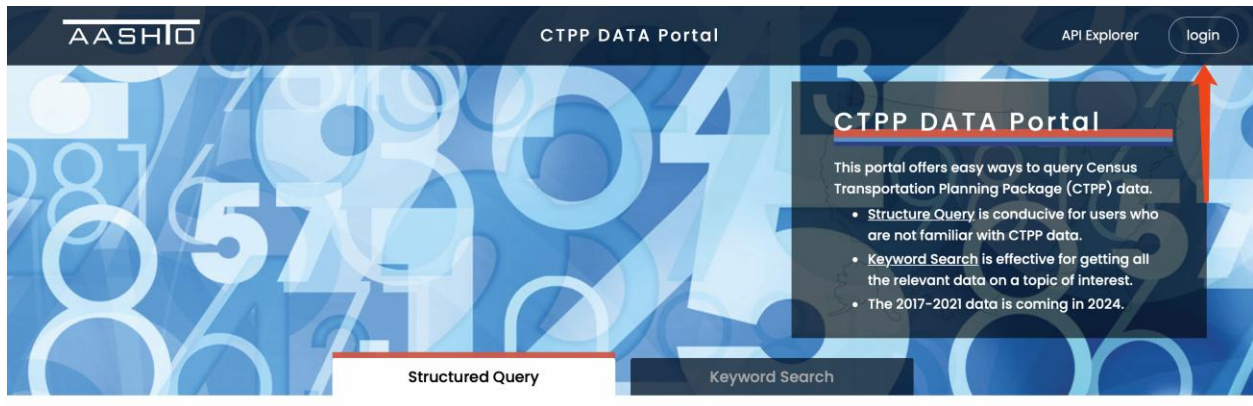
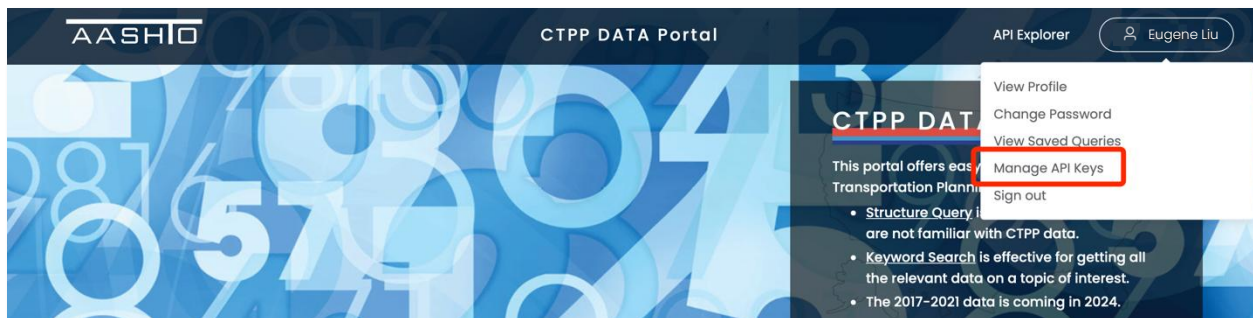


Figure 27. CTPP Data Portal Homepage

- Once login is successful, select the username button (which has replaced the “Login” button) to access the user action dropdown. Then, select the “Manage API



Keys” option.

Figure 28. Manage API Keys Option in the User Action Menu

- Select the “Generate New Key” button to create an API key.

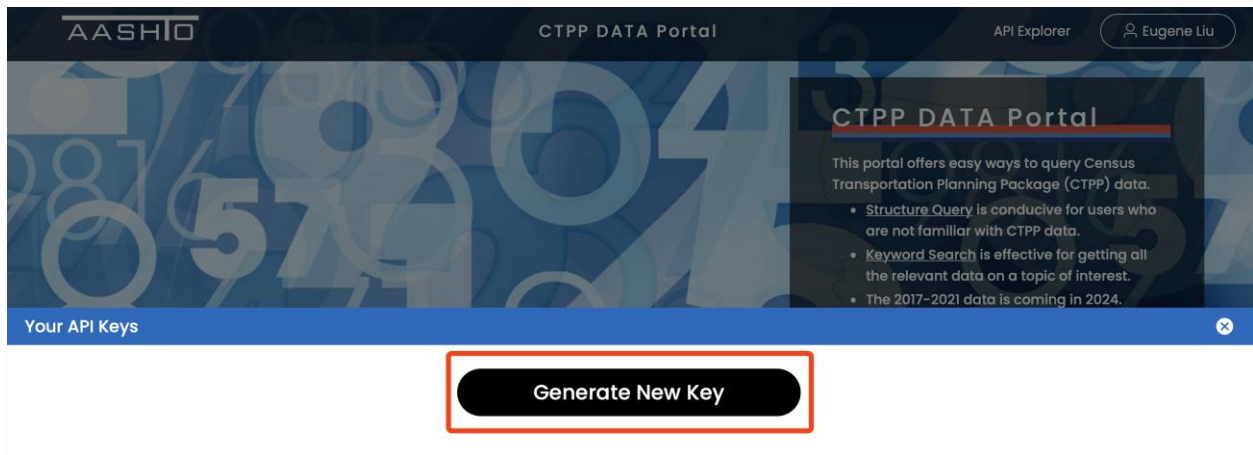


Figure 29. Generate New Key Button

- The new API key will appear in your API key list. This is the key you will use in the API Explorer and in your application code.

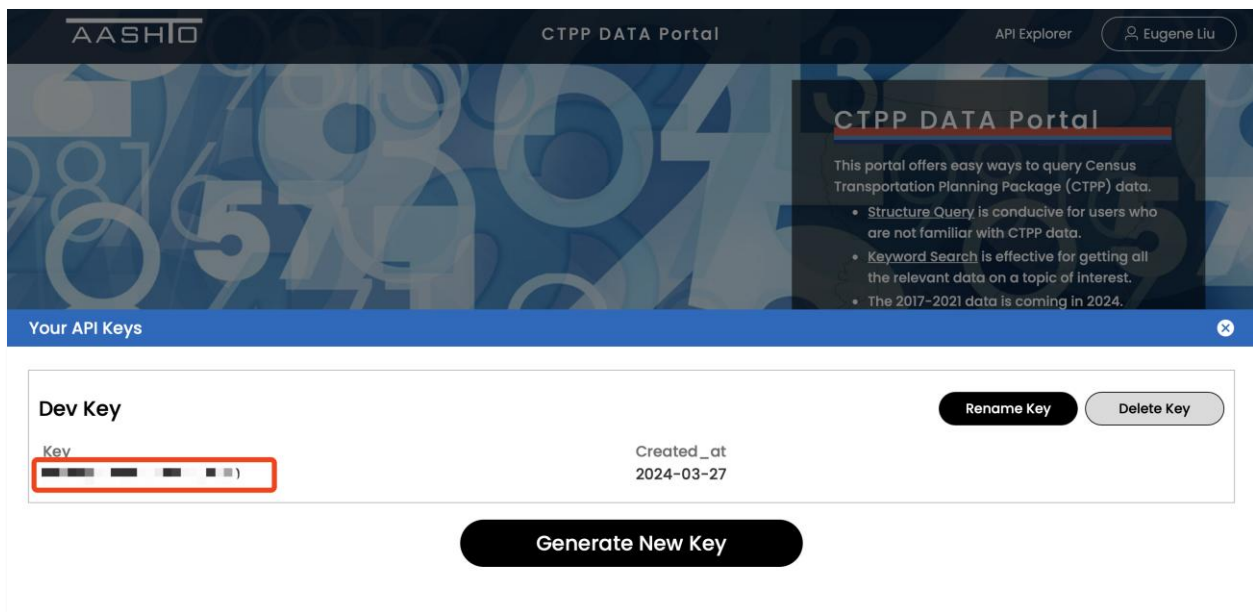
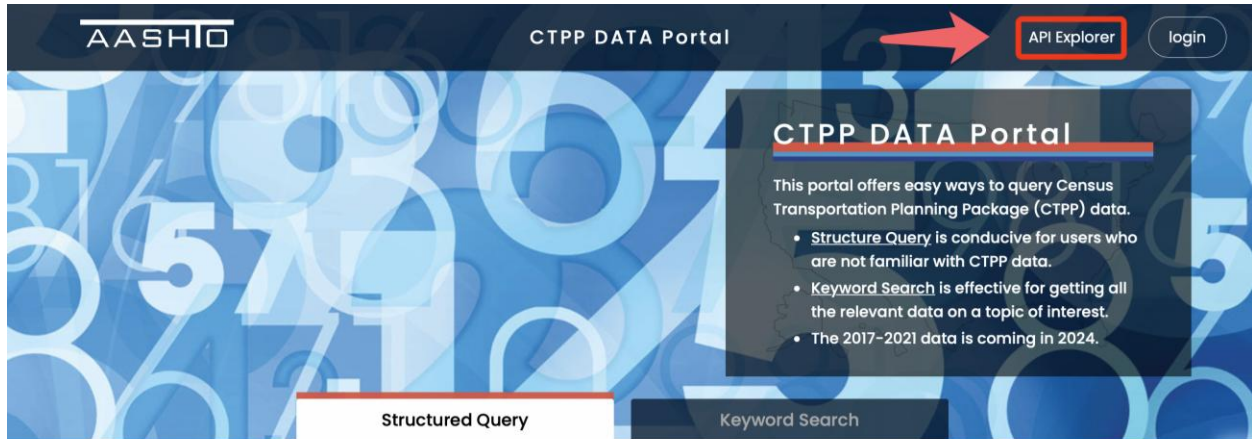


Figure 30. API Key List

Access the API Explorer

The API Explorer can be accessed one of two ways:

1. Via direct link: <https://ctppdata.transportation.org/api-explorer.html>
2. From the CTPP Data Portal, select the “API Explorer” button from the top navigation bar.



Use the API Explorer Functions

The API Explorer offers six key functions:

- Request authorization
- Endpoint navigation
- Method selection
- Parameter input
- Request execution
- Response handling

Each of these functions is described in the following sections.

1. **Request Authorization.** Before making any requests, you need to authorize your session by selecting the “Authorize” button from the API homepage (as shown in Figure 31), then entering the API key in the “Value” field and selecting the “Authorize” button in the pop-up window (as shown in Figure 32).

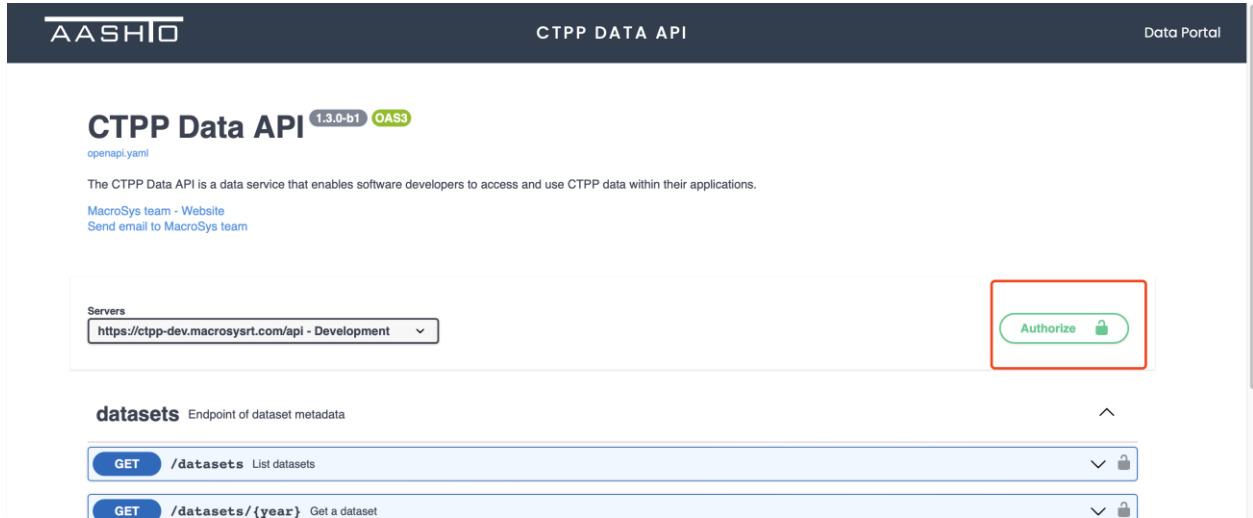


Figure 31. Authorize Button on API Homepage

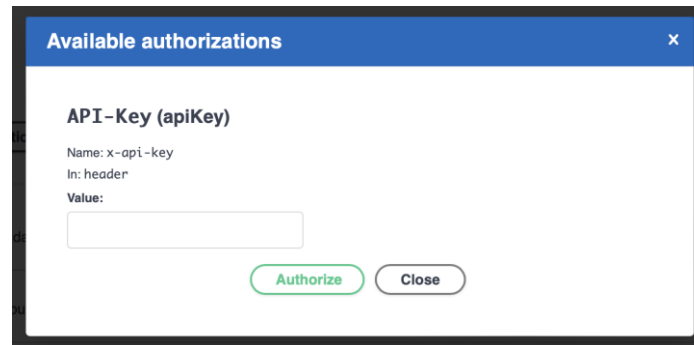


Figure 32. Value Field for API Key Entry to Authorize

2. **Endpoint Navigation:** Upon accessing the API Explorer, you will see a list of available endpoints grouped by category (as shown in Figure 33). Select an endpoint to expand it and view its details, including supported HTTP methods. The endpoints are grouped as follows: datasets, groups, and data.

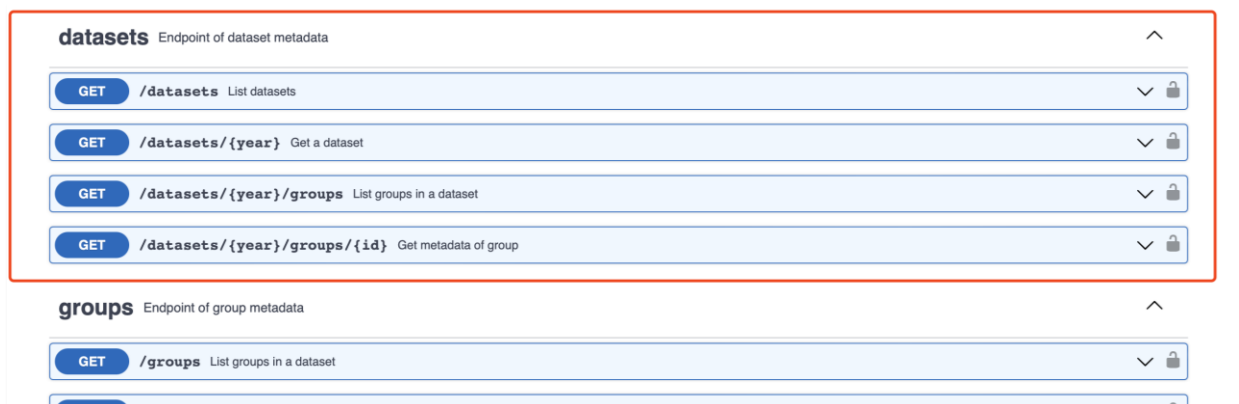


Figure 33. Available Endpoints Grouped by Category

3. **Method Selection:** Once you've expanded an endpoint, you will see the supported HTTP methods (as shown in Figure 34). Select a method to view its parameters, request body (if applicable), and responses.

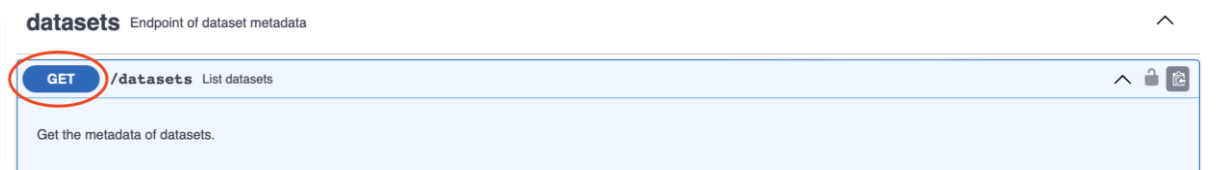


Figure 34. "Get" Button to Select a Method

4. **Parameter Inputs:** Parameters may include query parameters, path parameters, headers, or request body parameters. Fill in the required parameters with appropriate values (as shown in Figure 35, then select the "Try it out" button. The year parameter textbox will become editable. Set the "year" field value to one of the CTPP dataset years, such as 2016.

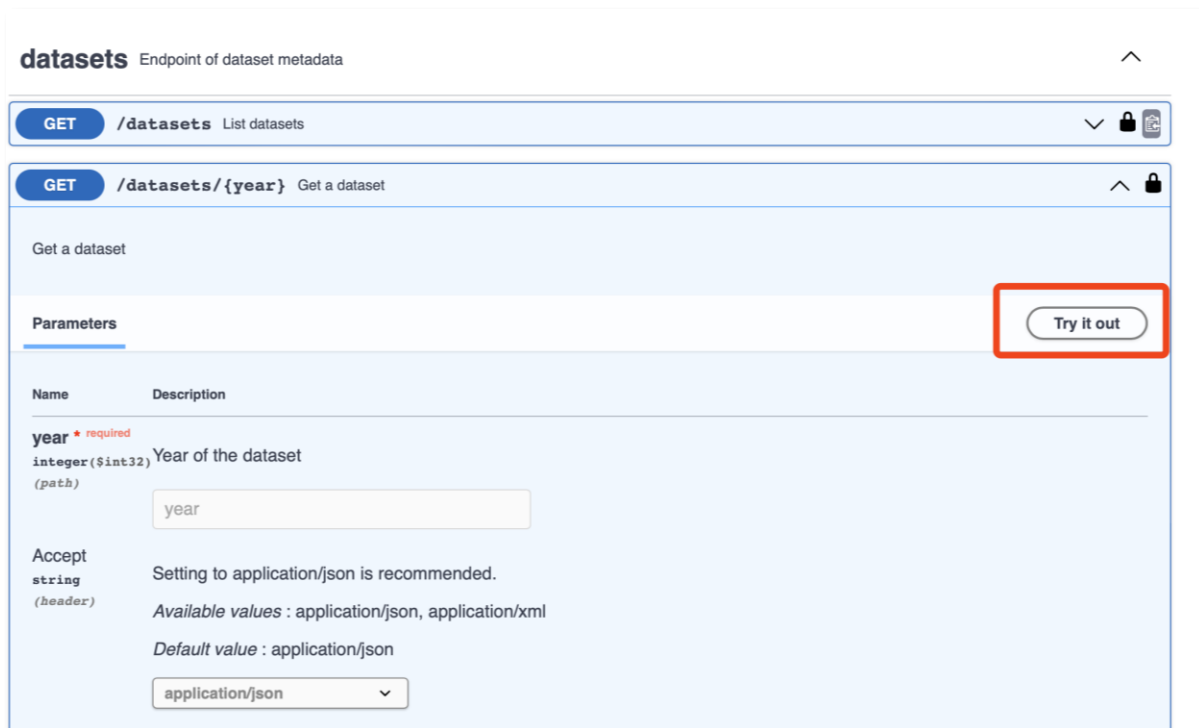


Figure 35. Parameter Inputs

5. **Request Execution:** After inputting parameters, select the "Execute" button to submit the request (as shown in Figure 36). The API Explorer will send the request to the API server and display the response below the parameters, including the response status code, headers, and body.

GET /datasets/{year} Get a dataset

Get a dataset

Parameters Cancel

Name	Description
year * required integer(\$int32) (path)	Year of the dataset

2016

Accept
string
(header)

Setting to application/json is recommended.

application/json

Execute

Figure 36. Request Execution

- Response Handling:** The API Explorer provides options to format and view the response body in JSON (as shown in Figure 37). You can also collapse or expand sections of the response for easier readability.

Responses

Code	Description	Links
200	A object of dataset metadata	No links

Media type: application/json

Examples: 2016

Controls Accept header.

Example Value | Schema

```
{
  "year": 2016,
  "title": "2012-2016 (5-year)",
  "description": "2016 dataset",
  "source": "U.S. Census Bureau, American Community Survey 2012-2016 Five-year estimates. Special Tabulation: Census Transportation Planning"
}
```

Figure 37. Option to Format and View Response Body